

YOUSSEF YASSER ELSAYED HAMED

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github.com/itz-youssef [Portfolio](#) [Kaggle](#) 📍 Giza, Egypt 12513

PROFILE

Third-year CS and AI student at Cairo University specializing in Machine Learning, Deep Learning, and Computer Vision, with hands-on experience building end-to-end ML systems for real-world applications. Strong in problem-solving, model optimization, and developing reliable AI solutions that are practical, scalable, and production-ready.

EDUCATION

Cairo University

Giza, Egypt

Bachelor's in Computer Science and Artificial Intelligence

Sep. 2023 – Sep. 2027

- **Relevant Coursework:** Algorithms Analysis & Design, Data Structures, Supervised Learning, Theoretical Foundations of Machine Learning, Autonomous Multiagent Systems, Object-Oriented Programming, Probability & Statistics, Computer Organization & Architecture, Math 1, Math 2, Math 3, Signals & Systems

EXPERIENCE

AI Engineer Intern

April 2026 – Present

Aitronix

On-site, New Cairo, Egypt

- Engineered a real-time **Behavioral AI Theft Detection** pipeline using **YOLOv11** + **ByteTrack** + **LSTM** to classify suspicious human-object interactions across 30 high-value target classes in retail environments.
- Maintained production-level code quality through modular architecture, peer code reviews, and iterative model improvements within a professional on-site engineering team.

Machine Learning Intern

April 2026

Code Alpha

Remote

- Delivered two production-ready classifiers under real deadlines — a **47-class** handwritten character recognizer (131K+ images, **94%** accuracy) and a binary tumor malignancy detector (560+ instances, **96%** accuracy).

Competitive Programmer

July 2024

Egyptian Collegiate Programming Contest (ECPC)

Alexandria, Egypt

- Engineered optimized algorithmic solutions in **C++** under strict time constraints, applying advanced data structures to tackle complex computational problems.
- Collaborated in a high-pressure team environment to prioritize problem selection, rapidly debug logical errors, and maximize competitive performance.

PROJECTS

YAI — Video Safety Analyzer [GitHub] | [Live Demo](#) | *PyTorch FastAPI, YOLOv10x, CLIP, ByteTrack* May 2026

- **End-to-End CV Pipeline:** Engineered a full-stack real-time video safety analysis system with a **FastAPI** backend and **Netlify**-hosted frontend, processing uploaded footage through a **5-stage** inference pipeline — detection, classification, tracking, anomaly scoring, and depth estimation — returning a structured JSON report.
- **Multi-Model Inference Stack:** Integrated **YOLOv10x** + **CLIP** open-vocabulary classification for frame-level detection, **ByteTrack** + **Kalman filtering** for persistent multi-person tracking across occlusions, **AnomalyTransformer** for behavioral anomaly scoring across **6** threat categories (loitering, crowd surge, forbidden zone, speed anomaly, trajectory anomaly, disappearance), and **Depth Anything v2** for monocular depth estimation.
- **Production-Grade Backend:** Built an async **FastAPI** server with **WebSocket** streaming for real-time frame-level result delivery, job queue management, and a **REST** polling API tunneled via a permanent **ngrok** domain with **zero-downtime** frontend decoupling between Netlify and the GPU backend.

Adaptive Pose & Action Intelligence System [GitHub] | *Python, PyTorch, YOLOv8, OpenCV, SQLite* May 2026

- **Multi-Level Human Analysis Pipeline:** Engineered a real-time computer vision system simultaneously tracking **527** anatomical points across 3 analysis levels: **17** body keypoints via YOLOv8-Pose, hand state estimation, and **68** facial landmarks via YuNet + LBF — achieving **55 FPS**.
- **Action & Posture Intelligence:** Extracted **8** joint angles per frame to compute real-time posture scores (0–100) and classify **6** body actions using a hybrid rule-based + PyTorch **LSTM** pipeline operating on 30-frame temporal sequences.

- **3-Layer Memory System:** Implemented a persistent behavioral memory using short-term RAM deque, unsupervised pattern learning across sessions, and a **SQLite** long-term store enabling cross-session posture trend tracking and anomaly flagging with threaded face analysis for optimal throughput.

Behavioral AI Theft Detection System [GitHub] | *Python, PyTorch, YOLOv11, SAHI, OpenCV* April 2026

- **Real-Time Vision Pipeline:** Engineered a computer vision pipeline using **YOLOv11x** for high-fidelity pose estimation and instance segmentation, actively monitoring **30** high-value target classes.
- **Robust Detection & Tracking:** Integrated **SAHI** (1000×1000 tiling) to maximize small-object recall, with **ByteTrack** and a **ResNet18**-based ReID manager for persistent multi-target tracking across occlusions.
- **Behavioral Classification:** Extracted **17** dynamic spatial features (grab motion, joint angles) to train a custom PyTorch **LSTM** network with an attention mechanism for classifying suspicious temporal movements.

Smart Agriculture Plant Disease Detection [Kaggle] | *PyTorch, TensorFlow/Keras, Gradio* Feb - May 2026

- **Automated Diagnosis:** Built an end-to-end plant disease classification system trained on **54,000+** PlantVillage images across **15** disease categories, benchmarking **14 models**(CNN (VGG-style), MobileNetV2, Vision Transformer, LSTM, Attention-CNN (CBAM), Random Forest, Bagging, KNN, GRU, AdaBoost, Autoencoder, Naive Bayes, Simple RNN, and EfficientNetB0) across Classical ML, Deep Learning, Transfer Learning, Sequential, and Advanced architectures.
- **Best Model CNN (VGG-style):** Designed and trained a custom **4-block VGG-style CNN** in PyTorch achieving **97.5%** validation accuracy, outperforming MobileNetV2 (**93.6%**), Vision Transformer (**93.3%**), and LSTM (**91.1%**).
- **Advanced Architectures:** Fine-tuned MobileNetV2 & EfficientNetB0 via transfer learning; implemented a custom Vision Transformer (ViT) with patch embeddings; built Attention-CNN with CBAM achieving **89.2%**; trained a DCGAN for rare-class augmentation and integrated Grad-CAM for disease localization.
- **Deployment:** Deployed best model as a Gradio web app with real-time prediction, confidence scores, and a treatment recommendation engine covering all **15** disease classes.

CERTIFICATES & AWARDS

Deep Learning Specialization <i>DeepLearning.AI (Andrew Ng) – Coursera</i>	May. 2026
Intermediate Machine Learning <i>Kaggle</i>	March 2026
Intro to Deep Learning <i>NVIDIA</i>	Dec. 2025
Egyptian Collegiate Programming Contest (ECPC) Participant <i>ICPC Foundation</i>	July 2024

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, Javascript

Developer Tools: Git, GitHub, Jupyter Notebook, Kaggle, Google Colab, SQLite, Docker

Machine Learning & AI: Deep Learning, Computer Vision, Predictive Modeling, Object Detection, Pose Estimation, Instance Segmentation, Anomaly Detection, Depth Estimation, Action Recognition, Behavioral Analysis, Transfer Learning, Attention Mechanisms, Hyperparameter Tuning, Transformer Models

Computer Vision & Perception: Multi-Object Tracking, Re-Identification (ReID), Optical Flow, Monocular Depth Estimation, Facial Landmark Detection, Hand State Estimation, Skeletal Keypoint Extraction, Open-Vocabulary Classification, Small-Object Detection (SAHI)

Data Science & Analysis: Exploratory Data Analysis (EDA), Feature Engineering, Dimensionality Reduction (PCA), Data Cleaning, Data Visualization, Data Augmentation, Statistical Analysis, Model Evaluation (ROC-AUC, Precision-Recall, Confusion Matrix, F1-Score)

Libraries & Frameworks: PyTorch, TensorFlow/Keras, Scikit-Learn, OpenCV, FastAPI, YOLO (v8, v10x, v11), CLIP, ByteTrack, Kalman Filter, SAHI, MediaPipe, Pandas, NumPy, Matplotlib, Seaborn, Moondream

Deployment: GPU Inference, ngrok Tunneling, Netlify, Vercel App, End-to-End ML System Design, Model Optimization, Pipeline Architecture

Software Engineering: Data Structures & Algorithms, Object-Oriented Programming, WebSocket Streaming, REST API Design, Fundamentals of Backend Development

Soft Skills: Problem-Solving, Critical Thinking, Team Collaboration, Time Management, Adaptability, Professional Communication

LANGUAGES

Arabic: Native or bilingual proficiency

English: Professional working proficiency